Semantic Prioritization of Domain Data for Proactive Identification of Cybersecurity Indicators

Nic Recasens & Nigel Smith

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Overall Status Update

- **Existing Literature**
  - Current systems struggle with novel threats, detection accuracy degrades rapidly without prior threat data.
  - There needs to be a shift from manual data engineering to semantic LLM reasoning.
  - Little has been done to empower the end user and focus on easily interpretable outputs

- **Our Team's Deliverables**
  - With that, our team looks mine semantic relevance from previously untapped raw data to enrich previous ML developments, not replace existing ML
  - Such enrichments could prevent more novel threats & improve human readability

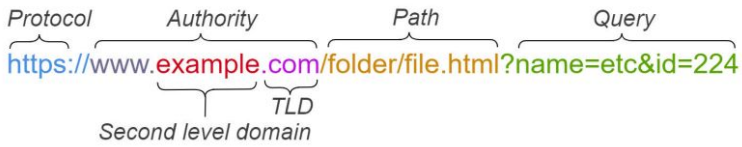


FIGURE 4. **Example of a URL with its components.**

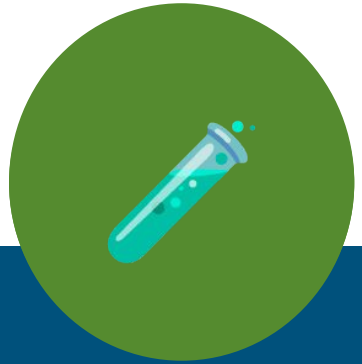
Research Questions Recap

Wide Questioning

Is there more data to be mined from the individual datum that is a phishing domain?

New Toolsets

What more can be done with these foundational domain data now that Natural Language Processing (NLP) has grown so rapidly in so few years?



Data Science POV

Is there a correlation between domain's lexical and semantic metadata?

Primary RQ

Do domain semantics help predict real-world phishing and impersonation activity to a statistically significant degree?

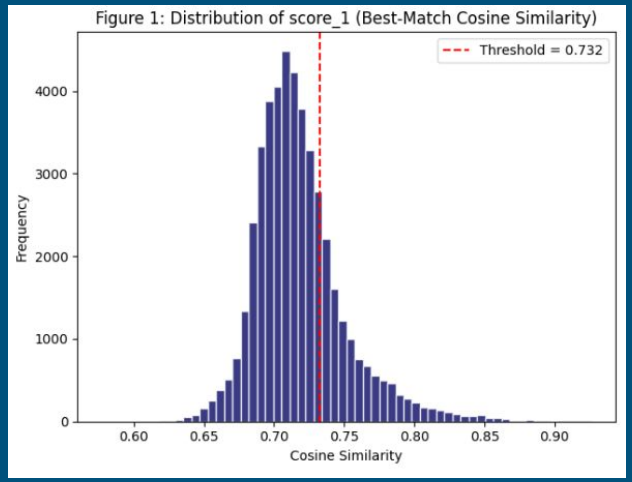
Dataset Description

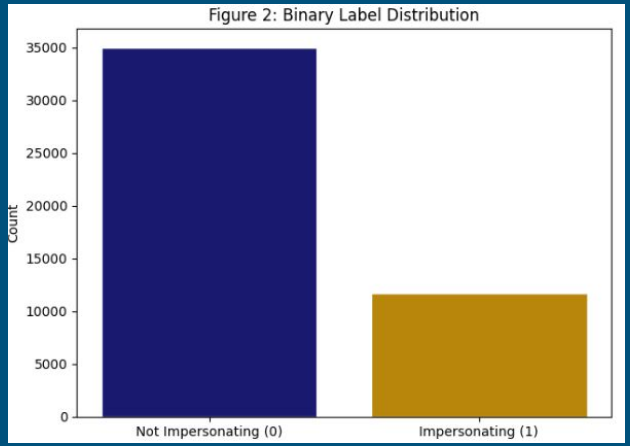
Sources: Cisco Phishtank Phishing URLs and Tranco Top 1 Million Domains

Preprocessing: URL cleaning, feature extraction, outlier removal

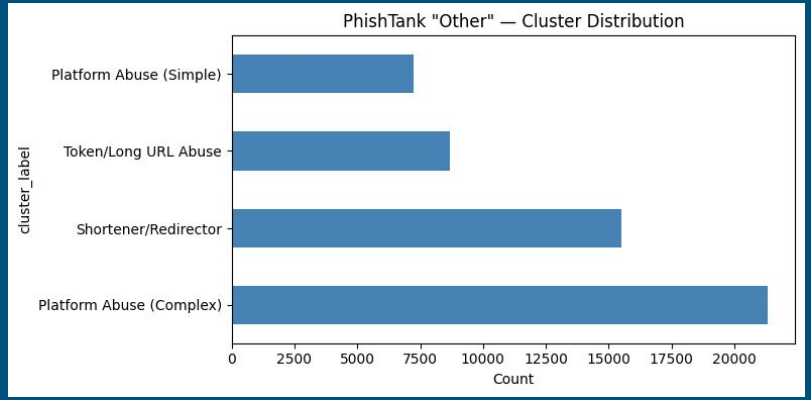
Key Features: `top-level domain`, `domain_length`, `dot_count`, `hyphen_count`, `digit_count`, and `digit_ratio`

Target Variables: `Phishing Target` and `Is Impersonation?`





Clustering



K-Means clustering applied to standardized lexical features to find common structures

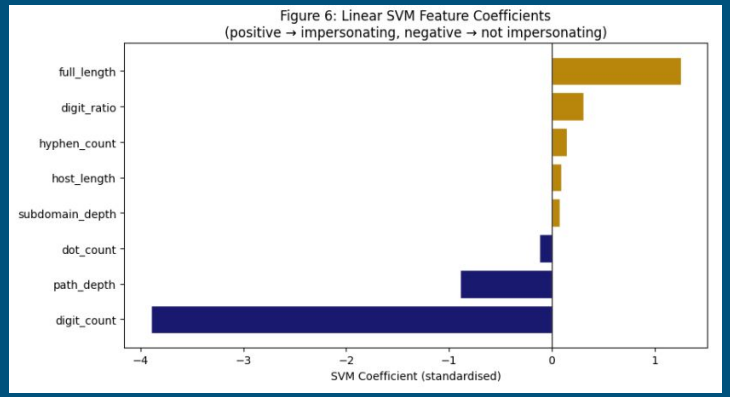
Elbow method applied to choose optimal number of categories to avoid introducing noise or overfitting

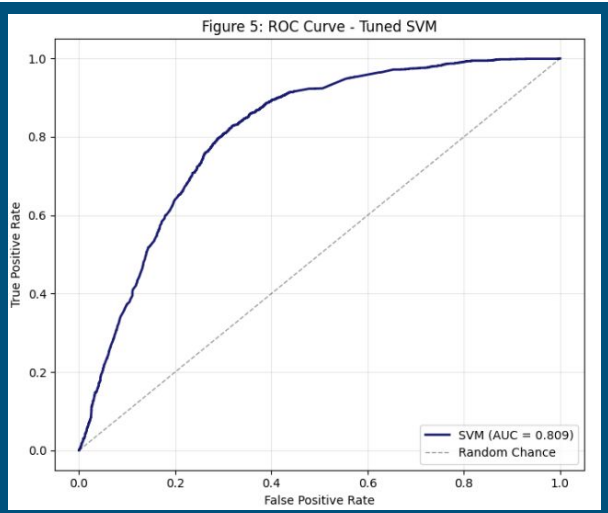
Platform abuse (simple and complex) is themost common: Google Sites, Wix, Weebly

Link shorteners: Data agrees that it is correct to be suspicious of them.

Google Docs, Dropbox: Abuse via long token strings

SVM





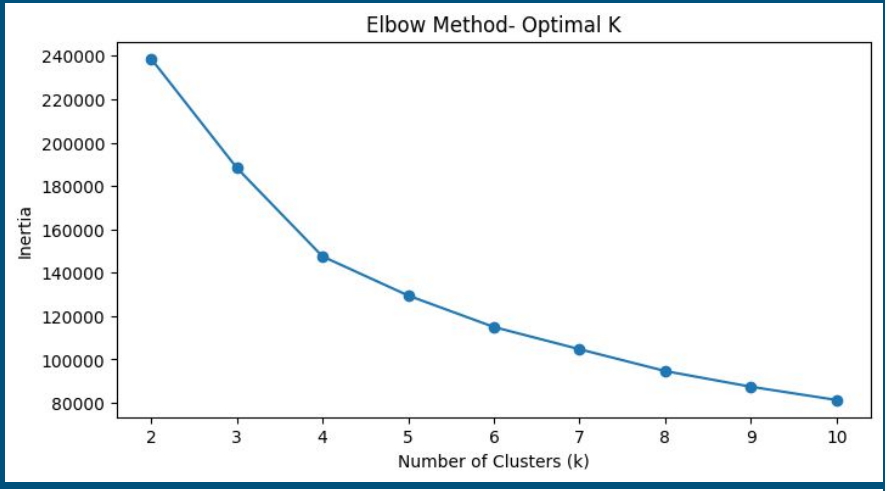
Baseline vs Tuned SVM - Hyperparameter tuning via grid search improved our default-parameter baseline, particularly in F1 score for the minority (impersonating) class.

ROC / AUC - Our AUC above 0.5 confirms the SVM is learning a signal beyond random chance from lexical URL features correlation.

Feature Importance - The linear SVM coefficient chart reveals which structural features most strongly push the prediction toward or away from the impersonating class.

Interpretation

Box plots show that impersonating URLs tend to have shorter overall lengths, fewer path segments, and lower subdomain depth than non-impersonating phishing URLs, whereas less targeted phishing URLs often rely on long, obfuscated paths.
An AUC above 0.5 confirms the SVM is learning a signal beyond random chance from lexical URL features alone, and the clusters match the proportion of phishing types we see discussed in our papers [4].

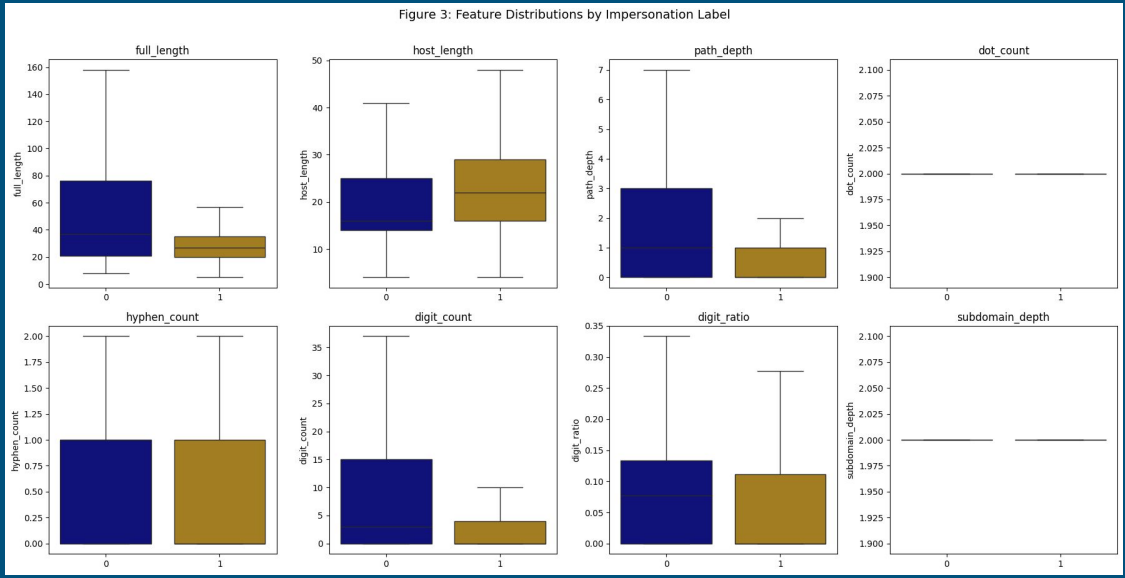


Cluster	Example URL
Platform Abuse (Simple)	https://pub-d637c8d427db484db2245748126bd788.r2.dev/rcubic.h...
Platform Abuse (Simple)	https://reserved-team-373417.framer.app/
Platform Abuse (Simple)	https://pub-cec2c769e7424225875847093b63e672.r2.dev/index.ht...
Platform Abuse (Complex)	https://please-wait--connection.firebaseapp.com/
Platform Abuse (Complex)	https://wxoontreonr.blogspot.com/
Platform Abuse (Complex)	https://startus-home.ghost.io/live-enus/
Token/Long URL Abuse	https://docs.google.com/presentation/d/e/2PACX-1vSd2kgD9d1Ub...
Token/Long URL Abuse	http://zuza.stefek.szczecin.pl/Webmail/75/Webmail/webmail.ph...
Token/Long URL Abuse	https://docs.google.com/presentation/d/e/2PACX-1vTMeGlpqWyMI...
Shortener/Redirector	https://qrco.de/bgbcV6
Shortener/Redirector	https://i8.ae/cHKFq
Shortener/Redirector	https://tinyurl.com/3xht4cub

```
AUC: 0.8093
Linear SVM - Test Set Results
                  precision    recall  f1-score   support

Not Impersonating      0.92      0.46      0.61      7008
    Impersonating      0.35      0.88      0.50      2336

         accuracy                           0.56      9344
        macro avg      0.63      0.67      0.55      9344
     weighted avg      0.78      0.56      0.58      9344
```



Reflections

Successes

- Structural URL features carry measurable data for predicting whether a phishing URL closely mimics a legitimate domain in embedding space.
- Labelled Phishing data is hard to come by, but using semantic reasoning to mine data is effective.
- A classic SVM classifier with hyperparameter tuning can achieves meaningful discrimination with our data.

Lessons Learned & Next Steps

- The impersonation label is derived from an embedding-based cosine similarity threshold, not from human annotation.
- PhishTank and Trancos both are not perfect data sources.
- The 75th percentile threshold should be refined.

Possible future work includes incorporating additional feature construction, using other classifiers, and pulling in new and varied datasets.

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